

University of Colorado at Colorado Springs
Department of Mechanical and Aerospace Engineering

SPCE 5065
HOMEWORK #4

DUE: Friday 17 July 2026

Instructions:

- Include your references at the end of your homework in AIAA format.
<https://www.aiaa.org/publications/journals/reference-style-and-format>
- Include units in your final answers!
- Write non-mathematical answers in complete sentences.
- Clearly state your assumptions.

2. In plasma physics, a Debye length is a measure of a charge carrier's net electrostatic effect in a plasma and how far its electrostatic effect persists.

a. Determine the Debye length in the lower ionosphere at an altitude of 300 km, where the electron temperature is approximately 1500 K. Assume a daytime solar max plasma density.

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T}{2ne^2}}$$
$$\lambda_D = \sqrt{\frac{(8.8541878 \times 10^{-12} \frac{A^2 s^4}{kg m^3})(1.380649 \times 10^{-23} \frac{J}{K})(1500K)}{2(5 \times 10^{12} \frac{1}{m^3})(1.60 \times 10^{-19} C)^2}}$$
$$\lambda_D = 846 \mu m$$

b. Determine the Debye length in the ionosphere at an altitude of 1000 km, where the electron temperature is approximately 5000 K. Assume a daytime solar max plasma density.

$$\lambda_D = \sqrt{\frac{(8.8541878 \times 10^{-12} \frac{A^2 s^4}{kg m^3})(1.380649 \times 10^{-23} \frac{J}{K})(5000K)}{2(5 \times 10^{12} \frac{1}{m^3})(1.60 \times 10^{-19} C)^2}}$$
$$\lambda_D = 15.5 \mu m$$

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3. An electromagnetic signal traverses the ionosphere vertically along a path that has a total electron content of 10^{18} electrons/m².

a. Determine the vertical time delay for a frequency of 150 MHz.

$$\Delta t_g \approx 40.31 \frac{TEC}{cf^2}$$

$$\Delta t_g \approx 40.31 \, m^3 Hz^2 \frac{\frac{10^{18}}{m^2}}{(3 \times 10^8 m/s)(150000000 \, 1/s)^2}$$

$$\Delta t_g \approx 5.98 \times 10^{-6} \, s$$

$$\Delta t_g \approx 5.98 \mu s$$

b. Determine the excess range for 150 MHz if the vacuum speed of light were used.

$$\Delta r = c \Delta t_g$$

$$\Delta r = \left(3 \times 10^8 \frac{m}{s} \right) (5.98 \times 10^{-6} s)$$

$$\Delta r = 1791.6 \, m$$

c. Determine the time delay if the frequency were 1.6 GHz.

$$\Delta t_g \approx 40.31 \, m^3 Hz^2 \frac{\frac{10^{18}}{m^2}}{(3 \times 10^8 m/s)(1.6 \times 10^9 /s)^2}$$

$$\Delta t_g \approx 5.25 \times 10^{-8} \, s$$

d. Determine the excess range for 1.6 GHz if the vacuum speed of light were used.

$$\Delta r = \left(3 \times 10^8 \frac{m}{s} \right) (5.25 \times 10^{-8} s)$$

$$\Delta r = 15.7 \, m$$

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4. Find an online ionospheric model. Provide an example of the model and describe it. Include in your description, at a minimum, who publishes it, where the data is gathered from, and any limitations of the model.

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5. Spacecraft in high altitude orbits are especially susceptible to spacecraft charging and can attain high potentials. Because the plasma is at a high temperature, the average speeds of the protons and ions are very large.

If we wish to minimize the induced current in a spherical geostationary satellite, how large will the spacecraft voltage be? The plasma temperature is 10^7 K.

Clearly state any assumptions you make.

The first order currents generated by the plasma environment for a spherical satellite are:

For electrons:

$$I_e = I_{e,o} A_e e^{\frac{eV}{k_B T_e}} \text{ for } V < 0 \text{ (repelled)}$$

$$I_e = I_{e,o} A_e \left[1 + \frac{eV}{k_B T_e} \right] \text{ for } V > 0 \text{ (attracted)}$$

Where:

$$I_{e,o} = \frac{1}{4} e n_e v_{mean}$$

$$v_{mean} = \sqrt{\frac{8k_B T_e}{\pi m_e}} - v_{spacecraft}$$

Where:

e = electron charge, C

V = voltage (V)

k_B = Boltzmann's constant, J/K

T_e = electron temperature, K

n_e = electron number density 1/m³

v_{mean} = relative velocity of electrons to spacecraft velocity (km/s)

For ions:

$$I_i = I_{i,o} A_i e^{\frac{eV}{k_B T_i}} \text{ for } V > 0 \text{ (repelled)}$$

$$I_i = I_{i,o} A_i \left[1 + \frac{eV}{k_B T_i} \right] \text{ for } V < 0 \text{ (attracted)}$$

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Where:

$$I_{i,o} = \frac{1}{4} e n_i v_{mean}$$

$$v_{mean} = \sqrt{\frac{8k_B T_i}{\pi m_i}} - v_{spacecraft}$$

Where:

T_i = ion temperature, K

n_i = ion number density $1/m^3$

v_{mean} = relative velocity of ions to spacecraft velocity (km/s)

a. What is the speed of the spacecraft?

Assume geostationary so orbit is circular.

$$V_{spacecraft} = \sqrt{\frac{398600.5}{42164}} = 3.075 \text{ km/s}$$

b. What are the mean speeds of the ions and electrons?

Assume:

$T_e = T_i = T_{plasma}$

Ion density = proton density

For electrons:

$$v_{e,mean} = \sqrt{\frac{8k_B T_e}{\pi m_e}} = \sqrt{\frac{8 * 1.3806505 * 10^{-23} * 10^7}{\pi * 9.1093826 * 10^{-31}}} = 19,645 \frac{km}{s}$$

For ions:

$$v_{i,mean} = \sqrt{\frac{8k_B T_e}{\pi m_i}} = \sqrt{\frac{8 * 1.3806505 * 10^{-23} * 10^7}{\pi * 1.66053873 * 10^{-27}}} = 460 \frac{km}{s}$$

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c. Find expressions for the induced potential currents for the electrons and ions in terms of V (spacecraft voltage). State your assumptions.

Assume:

Negative bias

$n_e \sim n_i A_e = 2 A_i$

$A_e \sim 2 A_i$

$$I_{e,o} = \frac{1}{4} e n_e v_{mean}$$

$$I_{e,o} = \frac{1}{4} e n_e * (19645 - 3.075) * 1000 \frac{Amps}{m^2}$$

$$I_e = I_{e,o} A_e e^{\frac{eV}{k_B T_e}} \text{ for } V < 0 \text{ (repelled)}$$

$$I_e = \frac{1}{4} e n_e * (19645 - 3.075) * 1000 * 4 * A * e^{\frac{eV}{k_B T_e}} Amps$$

$$I_{i,o} = \frac{1}{4} e n_i v_{i,mean}$$

$$I_{i,o} = \frac{1}{4} e * n_i (460 - 3.075) * 1000 \frac{Amps}{m^2}$$

$$I_i = I_{i,o} A_i \left[1 - \frac{eV}{k_B T_i} \right] \text{ for } V < 0$$

$$I_i = \frac{1}{4} e n_i * A * \left[1 - \frac{eV}{k_B T_i} \right] Amps$$

d. Find an expression in terms of V for total current flow.

$$I = (19645 - 3.075) * 2 * e^{\frac{eV}{k_B T_e}} - (460 - 3.075) * \left[1 - \frac{eV}{k_B T_i} \right] Amps = 0$$

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e. For no current flow, what would the spacecraft voltage be?

Solve numerically, in Matlab solution is (see attached code)

$V = -2626$ Volts

f. Is this a high risk to the spacecraft? If so, what would you recommend to reduce the risk?

No, you could bias the spacecraft near this induced voltage to reduce the probability of a discharge event.

6. One of the reasons most earth orbiting spacecraft are negatively biased is due to the more widespread use of npn versus pnp transistors.

a. What are npn and pnp transistors and what are their advantages and disadvantages?

b. Why might they be of wider use on earth orbiting spacecraft than pnp transistors?

7. Select an article in a peer-reviewed journal or NASA document that describes the electrical grounding of spacecraft and summarize its contents.

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8. Determine the voltage with respect to its environment of a spacecraft at synchronous altitude if the plasma temperature is 1×10^7 K. Consider the environment to consist primarily of electrons and protons. Clearly state your assumptions.

Assume $A_i/A_e = 1/2$

$$V = \frac{k_B T_e}{e} \ln \left(\frac{4V_o \frac{A_i}{A_e}}{V_e} \right)$$

Where

$$v_{e,mean} = \sqrt{\frac{8k_B T_e}{\pi m_e}} = \sqrt{\frac{8 \cdot 1.3806505 \cdot 10^{-23} \cdot 10^7}{\pi \cdot 9.1093826 \cdot 10^{-31}}} = 19,645 \frac{km}{s}$$

$$V = \frac{(1.3806505 \cdot 10^{-23} \frac{J}{K})(1 \times 10^7 K)}{1.6 \times 10^{-19} C \cdot \frac{J/V}{C}} \ln \left(\frac{4 \sqrt{\frac{398600.5 \frac{km}{s}}{42164 \frac{s}{2}}}}{19645 \frac{km}{s}} \right)$$

$$V = 862.91 (-8.069234)$$

$$V = -6963 \text{ Volts}$$